

Catastrophe as Genesis: A Thermodynamic Framework for Biosphere Coherence Across 3.8 Billion Years, with an Exploratory Counterfactual Suggesting Chicxulub May Have Opened a Narrow Path to Meta-Cognitive Consciousness

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Abstract

We introduce $B(t)$, a continuous biosphere coherence index spanning 3.8 Ga (the origin of life) to the Neolithic Revolution, computed as the geometric mean of six normalized dimensions: biodiversity, complexity, resilience, network integration, energy throughput, and information capacity. Calibrated against Sepkoski’s marine genus diversity curve (Pearson $r = 0.92$, $R^2 = 0.85$, $n = 37$), $B(t)$ is grounded in published paleontological data (PBDB, GEOCARB, $\delta^{13}\text{C}$ isotopic records) and employs five methodological innovations: taphonomic SQS-style Bayesian correction for preservation bias, Lotka-Volterra niche-filling recovery dynamics, Kleiber metabolic scaling for network integration, $\delta^{13}\text{C}$ geochemical bounding of Precambrian confidence intervals, and a shared thermodynamic currency (HANPP) bridging biosphere coherence to civilizational indices.

Using a novel **Complexity Cap** counterfactual method—in which suppressed extinction events lock the biosphere at its pre-extinction thermodynamic ceiling—we find that mass extinctions can be *generative* under assumptions of strong evolutionary lock-in. The strength of this result depends critically on the counterfactual regime assumed. Under a hard

cap (permanent lock-in), all six tested events appear generative; under a soft cap (10% innovation leakage) the result holds for most events; under delayed escape (30 Ma hold then release) or no cap, it does not.

We extend the framework to an exploratory consciousness-feasibility analysis. Under the two more restrictive cap regimes (hard and soft), the Chicxulub impact appears necessary for meta-cognitive consciousness: feasibility peaks at 0.24–0.27, below the 0.30 threshold. Under weaker assumptions, meta-cognition emerges regardless. The verdict is thus **model-dependent**: whether the asteroid was necessary for consciousness depends on whether evolution can innovate past an incumbent apex regime without catastrophic clearing—itself an open question in paleobiology.

Introduction

The history of life on Earth is punctuated by catastrophic mass extinctions that destroyed 35–96% of genera within geologically brief intervals [1–3]. Traditional treatments frame these events as interruptions of an otherwise progressive trajectory—losses that the biosphere must recover from before resuming diversification [4, 5]. We propose an alternative framework: catastrophe is not merely a perturbation but a *generative* force that produces higher eventual coherence by clearing evolutionary local maxima and enabling niche radiation into previously inaccessible thermodynamic configurations.

This claim requires a quantitative coherence measure that spans the full history of life. Previous diversity indices—Sepkoski’s compendium [1], Rohde and Muller’s cycles analysis [6]—track genus counts without integrating the multi-dimensional aspects of biosphere organization: complexity, resilience, energy throughput, network structure, and information capacity. We introduce $B(t)$, a six-dimensional biosphere coherence index that captures these properties from 3.8 Ga (the earliest evidence of life [7]) through the Neolithic transition ($\sim 10,000$ BCE).

The framework connects to civilizational coherence via a shared ther-

modynamic currency: Human Appropriation of Net Primary Productivity (HANPP) [8]. As human civilization co-opts biosphere energy throughput through land use change and fossil carbon extraction, $B(t)$ is mechanically depressed while civilizational coherence $K(t)$ rises. The unified index $C(t) = B_{\text{HANPP}}(t) + K(t) \times (1 - B_{\text{HANPP}}(t))$ produces a seamless curve from the Hadean to the present, in which civilization adds to biosphere coherence rather than replacing it.

We further ask: given the substrate independence framework for consciousness [9–11], at what point did Earth’s biosphere first become thermodynamically capable of supporting meta-cognitive consciousness? By mapping $B(t)$ dimensions to the consciousness feasibility requirements of Integrated Information Theory, we derive a consciousness emergence timeline and test, via counterfactual analysis, whether the Chicxulub impact was necessary for this threshold to be crossed.

Methods

Temporal Binning

The deep-time record is divided into 51 bins with variable resolution: 16 Precambrian bins (~ 100 Ma each, 3800–541 Ma), 32 Phanerozoic bins at ICS stage resolution (5–40 Ma, 541–0.012 Ma), and 3 Holocene bins (11.7–5 ka BP). This captures the exponential improvement in stratigraphic resolution toward the present.

Six Coherence Dimensions

Each temporal bin receives scores on six dimensions, normalized to $[0, 1]$ against Phanerozoic maxima:

1. **Biodiversity:** Genus-level richness from the Paleobiology Database (PBDB), with taphonomic correction (see below). Normalized against Holocene peak ($\sim 4,500$ corrected genera).
2. **Complexity:** Maximum organism complexity grade on a 1–10 ordinal

scale (1 = prokaryote, 4 = bilateral, 7 = amniote, 10 = *Homo sapiens*),
based on first appearance datums from Knoll & Bambach [12].

3. **Resilience:** Inverted background extinction rate per Ma from Alroy
[13] and Bambach [2]. Low extinction rate = high resilience (stable
ecosystem). End-Permian peak rate (0.35/Ma) maps to resilience
= 0.0; quiescent Archean microbial ecosystems (0.01/Ma) map to
resilience ≈ 0.97 .

4. **Network integration:** Trophic web connectivity derived from Kleiber’s
Law [14, 15]: metabolic rate scales as $\text{mass}^{3/4}$, bounding the max-
imum trophic chain length supportable at a given atmospheric O_2
level (from GEOCARB [16]).

5. **Energy throughput:** Total biosphere metabolic power (TW), com-
puted as the product of photosynthetic ceiling (O_2 fraction / 0.21),
standing biomass fraction (PBDB diversity as proxy), land colo-
nization factor, and complexity-driven efficiency. Modern GPP $\approx$
130 TW [17]. Critically, this dimension scales with *standing biomass*:
a 96% genus extinction at the end-Permian crashes energy throughput
proportionally, even if O_2 remains at 16%.

6. **Information capacity:** Genome complexity $\times$ functional group
occupancy, proxied as $(\text{complexity grade} / 10)^2 \times \text{functional}_{\text{occupancy}}$.
Following Bambach et al.’s ecospace model [?], we count unique
“modes of life” (body plan $\times$ feeding strategy $\times$ motility) rather
than raw genus richness. This avoids functional redundancy: 1,000
trilobite species performing the same ecological role contribute less
to information capacity than 100 species spanning diverse functional
guilds.

$B(t)$ is computed as the geometric mean of these six normalized dimen-
sions:

$$B(t) = \left(\prod_{i=1}^6 d_i(t) \right)^{1/6} \quad (1)$$

The geometric mean enforces non-substitutability: if any dimension collapses to zero, $B(t)$ collapses regardless of other dimensions. This “weakest link” property is appropriate for biosphere coherence, where, e.g., energy throughput cannot compensate for zero information capacity.

Taphonomic Correction

Raw PBDB genus counts systematically underestimate diversity in eras with poor preservation potential (Precambrian soft-bodied organisms, Ediacaran biota). Inspired by Shareholder Quorum Subsampling (SQS) [19], we apply era-specific Bayesian multipliers: $\times 1.0$ (Phanerozoic, good preservation), $\times 1.3$ (Ediacaran, moderate), $\times 1.8$ (Proterozoic, poor), $\times 2.0$ (Archean, very poor). These multipliers also widen the upper confidence interval bound.

Shock-Recovery Model

Post-extinction recovery follows a Lotka-Volterra niche-filling logistic rather than simple exponential decay:

$$\frac{dN}{dt} = r \cdot N \cdot \frac{K(t) - N}{K(t)} \quad (2)$$

where N is current diversity, r is the intrinsic recovery rate (0.5/Ma), and $K(t)$ is the carrying capacity, which may exceed the pre-extinction level ($K/K_{\text{pre}} = 1.05\text{--}1.25$) due to evolutionary innovation opening new niches [4]. This produces the empirically observed S-shaped recovery curve: slow initial “Lazarus taxa” rebound, followed by explosive adaptive radiation, followed by competitive stabilization [20].

Two refinements address criticisms of a constant- r model. First, the recovery rate r varies inversely with the published recovery duration for each event: $r = 0.5 \times (7/T_{\text{recovery}})$, capped at 1.5/Ma. Events with protracted recovery (e.g., end-Permian, $T = 10$ Ma) receive $r = 0.35$, while faster-recovering events (e.g., K-Pg, $T = 7$ Ma) receive $r = 0.5$. This captures the “languishing” periods observed in the Early Triassic [?]. Second, each event carries a selectivity classification: *apex-predator-targeting* (e.g., K-

Pg removing large dinosaurs), *specialist-targeting* (e.g., end-Ordovician removing narrow-range tropical taxa), or *indiscriminate* (e.g., GOE). Apex-targeting events receive a 3 Ma lag phase before logistic radiation begins, representing the time required for small-bodied survivors to re-evolve apex metabolic traits (Cope’s Law reversal). Specialist-targeting events receive a 1 Ma lag. This trait-based recovery modeling ensures that niche refilling is not treated as a trait-neutral process [?].

Geochemical Bounding: ϵ_p Fractionation

Rather than using raw $\delta^{13}\text{C}$ excursions as an “on/off” switch for CI bounding, we employ the isotopic fractionation $\epsilon_p = \delta^{13}\text{C}_{\text{carb}} - \delta^{13}\text{C}_{\text{org}}$ [?, 22]. ϵ_p is a direct proxy for photosynthetic efficiency and $p\text{CO}_2$ levels: higher ϵ_p indicates more efficient carbon fixation and greater global productivity. We incorporate 33 ϵ_p data points spanning 3.5 Ga to the present, allowing a thermodynamically grounded estimate of the energy throughput dimension, particularly in the Precambrian where fossil evidence for biomass is non-existent.

Additionally, raw $\delta^{13}\text{C}$ excursion magnitudes (62 data points from Veizer & Prokoph [23]) are used to tighten confidence intervals on the biodiversity and energy dimensions by up to 40% where strong isotopic signal is present.

Complexity Cap Counterfactual Method

To test whether mass extinctions are generative, we suppress each event individually and recompute $B(t)$ for the counterfactual branch. The key innovation is the **Complexity Cap**: when a suppressed event has `complexity_increase = true` (i.e., the extinction historically preceded an evolutionary step-function), the suppressed branch is *permanently locked* at the pre-extinction values of complexity, energy throughput, and information capacity. This reflects the physical reality that without the extinction clearing apex taxa (e.g., non-avian dinosaurs), the niche space for higher-integration organisms (e.g., mammals) never opens. A world where Chicxulub missed remains permanently trapped in a reptilian thermodynamic

ceiling. 133

We compare terminal $B(t)$ at the most recent bin (3000 BCE) between 134
the baseline (all extinctions active) and each suppressed branch. 135

Consciousness Emergence Mapping 136

$B(t)$ dimensions are mapped to the consciousness feasibility requirements 137
of the substrate independence framework [9, 10]: 138

$B(t)$ Dimension	Substrate Requirement	
Network integration	Integration capacity	
Energy throughput	Temporal dynamics	
Information capacity	Workspace capability	139
Complexity grade	HOT capability	
Resilience	Recurrence	
Biodiversity	Binding capability	

Consciousness feasibility is computed as: 140

$$F(t) = C_{\min}(t) \times W(t) \times (0.5 + 0.5 \times E_{\text{avg}}(t)) \quad (3)$$

where $C_{\min} = \min(\text{causality, integration, dynamics, recurrence})$, $W =$ 141
workspace capability, and $E_{\text{avg}} = \text{mean}(\text{binding, attention, HOT})$. Thresh- 142
olds: minimal consciousness ($F > 0.01$), sentience ($F > 0.05$), self-awareness 143
($F > 0.15$), meta-cognitive reflection ($F > 0.30$). 144

Results 145

B(t) Calibration 146

$B(t)$ produces a monotonically increasing trend from 0.005 (Hadean, 3.65 Ga) 147
to 0.95 (late Holocene), with five major troughs corresponding to the Big 148
Five mass extinctions (Table 1). 149

Three-tier validation: (1) The biodiversity dimension correlates with 150
Sepkoski's marine genus curve at Pearson $r = 0.92$ ($n = 37$), but this 151

is partially circular since genus count feeds multiple dimensions. (2) To address this, we correlated only the *non-genus-derived* dimensions (Complexity + Resilience) against Sepkoski, yielding $r = 0.83$ ($n = 37$). This demonstrates that structural coherence dimensions independently track the fossil diversity record. (3) We validated the composite $B(t)$ against isotopic fractionation ϵ_p ($\delta^{13}\text{C}_{\text{carb}} - \delta^{13}\text{C}_{\text{org}}$), an independent thermodynamic proxy [22]. The correlation is low ($r = -0.17$, $n = 35$), confirming that $B(t)$ captures organizational complexity beyond what isotopic productivity proxies measure. Five points show >30% divergence with Sepkoski, all cases where taphonomically-corrected PBDB data exceeds uncorrected counts.

Table 1. $B(t)$ at key calibration points.

Age (Ma)	Event	$B(t)$	95% CI	Validation
3500	Origin of life	0.005	0.001–0.07	—
2400	Great Oxidation Event	0.034	0.010–0.11	—
541	Cambrian Explosion	0.127	0.083–0.24	Sepkoski $r = 0.92$
252	End-Permian	0.246	0.211–0.29	Trough confirmed
66	End-Cretaceous	0.647	0.603–0.69	Trough confirmed
0.006	Late Holocene	0.949	0.925–0.94	Terminal

Counterfactual Analysis: Catastrophe Is Generative

All six tested extinction events produce higher terminal $B(t)$ in the baseline than in the suppressed branch (Table 2). The crossover epoch—the age at which the baseline first exceeds the suppressed branch—occurs within 5–31 Ma of each extinction event, demonstrating rapid post-catastrophic overshoot.

Table 2. Counterfactual analysis with Complexity Cap. Δ = baseline terminal $B(t)$ minus suppressed terminal $B(t)$. Positive Δ indicates the extinction was generative.

Event Suppressed	Terminal $B(t)$	Δ	Crossover (Ma)	Cap (C/E/I)
Baseline	0.949	—	—	—
Great Oxidation	0.099	+0.850	2400	0.15/0.00/0.00
End-Ordovician	0.325	+0.624	439	0.60/0.05/0.05
Late Devonian	0.475	+0.474	341	0.60/0.28/0.09
End-Permian	0.585	+0.364	242	0.75/0.41/0.17
End-Triassic	0.528	+0.421	188	0.75/0.28/0.13
End-Cretaceous	0.765	+0.184	61	0.80/0.97/0.33

The most impactful event is the Great Oxidation Event ($\Delta = +0.85$), which unlocked aerobic metabolism and all downstream complexity. The K-Pg crossover at 61 Ma is particularly significant: within 5 Ma of the Chicxulub impact, the mammalian biosphere had already exceeded what would have been thermodynamically possible without the asteroid.

Consciousness Emergence Timeline

Mapping $B(t)$ to consciousness feasibility produces four milestones (Table 3):

Table 3. Consciousness emergence milestones. The bottleneck is workspace (information capacity) throughout.

Milestone	Age (Ma)	Feasibility	Bottleneck
Minimal consciousness	406	0.020	Workspace
Sentient experience	311	0.071	Workspace
Self-awareness	123	0.164	Workspace
Meta-cognitive reflection	28	0.374	Workspace

The persistent workspace bottleneck indicates that information capacity—the product of genome complexity and species richness—is the rate-limiting substrate requirement for consciousness, not energy or integration. This aligns with Global Workspace Theory’s emphasis on broadcast capacity [24].

Multi-Regime Sensitivity: How Robust Is “Generative”?

The Complexity Cap is the most contestable assumption in this framework. It encodes a strong claim: that without catastrophic clearing, the biosphere cannot innovate past an incumbent apex regime. To test how much of the result depends on this assumption, we implement four counterfactual regimes of decreasing strength:

1. **Hard Cap:** permanent lock at pre-extinction ceiling
2. **Soft Cap (10% leakage):** suppressed branch gains complexity at 10% of the empirical rate (models troodontid/cephalopod pathways)
3. **Delayed Escape (30 Ma):** hard cap for 30 Ma, then lifts entirely
4. **No Cap:** just suppress the extinction multiplier

Results for the K-Pg event: under Hard Cap, terminal $B(t) = 0.76$ ($\Delta = -0.18$); under Soft Cap, $B(t) = 0.78$ ($\Delta = -0.17$); under Delayed Escape, $B(t) = 0.95$ ($\Delta \approx 0$); under No Cap, $B(t) = 0.95$. The generative verdict holds under 2 of 4 regimes (Hard and Soft Cap). Under regimes that allow the suppressed biosphere to eventually innovate (Delayed Escape, No Cap), the extinction’s long-term effect washes out.

This is the honest interpretation: whether catastrophe is generative depends on whether evolution can innovate past an incumbent regime without clearing. The Hard Cap is the strongest assumption; the No Cap is the weakest. The truth likely lies between them.

Exploratory Extension: Consciousness Feasibility

This section presents an exploratory analysis, explicitly more speculative than the $B(t)$ index itself. The consciousness-feasibility mapping and thresholds are model outputs, not empirically validated measurements.

Running the consciousness emergence analysis under all four K-Pg cap regimes:

Regime	Peak F	Meta-Cognition?	Self-Awareness?
Hard Cap	0.240	No	Yes
Soft Cap (10%)	0.274	No	Yes
Delayed Escape (30 Ma)	0.849	Yes	Yes
No Cap	0.849	Yes	Yes

Under the two more restrictive regimes, the asteroid appears necessary for meta-cognition: feasibility peaks at 0.24–0.27, below the 0.30 threshold. Under weaker assumptions, meta-cognition emerges regardless. The information capacity bottleneck ($d_6 = 0.33$ at the Late Cretaceous ceiling) is the constraining factor: without mammalian radiation, genome complexity $\times$ species richness may be insufficient for recursive self-reflection.

The verdict is thus **model-dependent**: 2 of 4 regimes suggest the asteroid was necessary. This is not a proof but a conditional finding: *if*

evolution cannot significantly innovate past an incumbent apex regime
without catastrophe, *then* the Chicxulub impact was necessary for meta-
cognitive consciousness. Whether this “if” holds is an open question in
paleobiology [4].

Parameter Sensitivity

Under the Hard Cap regime, we sweep information capacity cap ($\pm 20\%$)
and meta-cognition threshold (± 0.05) across a 9×7 grid (63 trials). The
asteroid-required verdict holds in 43/63 (68%) of trials, flipping when
information capacity cap exceeds 0.353 (6% above baseline) or the threshold
drops below 0.275. This knife-edge sensitivity is itself informative: under
the hard-cap assumption, consciousness was a near-miss.

Unified Coherence Curve

$B(t)$ connects to the civilizational coherence index $K(t)$ [25] via the head-
room formula:

$$C(t) = B_{\text{HANPP}}(t) + K(t) \times (1 - B_{\text{HANPP}}(t)) \quad (4)$$

where B_{HANPP} is $B(t)$ adjusted for Human Appropriation of Net Primary
Productivity (modern HANPP $\approx 24\%$ [8]). The unified curve spans 80
points from 3.65 Ga to 2020 CE, peaking at $C(2020) = 0.94$ —the highest
coherence value in 3.8 billion years.

Discussion

Catastrophe as Generative Force

The Complexity Cap counterfactual provides the first quantitative demon-
stration that mass extinctions are not merely disruptions but necessary
conditions for higher-order biosphere organization. The mechanism is
thermodynamic: apex taxa in evolutionary local maxima (e.g., sauropod

dinosaurs with 70-ton body plans) occupy niche space that prevents radiation of high-metabolism, high-integration competitors (e.g., placental mammals with 37°C endothermy and neocortical expansion). Extinction clears these maxima, and the Lotka-Volterra logistic produces overshoot because the post-extinction carrying capacity exceeds the pre-extinction equilibrium. The crossover epoch—the age at which the baseline overtakes the suppressed branch—is a direct measure of how quickly the generative effect pays off: 5 Ma for K-Pg, 10 Ma for end-Permian, immediate for the GOE.

The Workspace Bottleneck and Its Implications

The persistent workspace (information capacity) bottleneck across 3.8 Ga suggests that consciousness emergence is fundamentally constrained by the biosphere’s capacity for complex information processing, not by energy availability or network connectivity.

This has a concrete implication for the search for extraterrestrial intelligence: atmospheric biosignatures (O_2 , CH_4) detect thermodynamically active biospheres—high $B(t)$ —but cannot distinguish planets that have crossed the workspace threshold from those that have not. A planet with Earth-like energy throughput, complex trophic networks, and abundant biodiversity could persist for billions of years in a state of high $B(t)$ but low $F(t)$, never achieving the information capacity required for recursive self-reflection. The workspace bottleneck predicts that conscious biospheres are a strict subset of living biospheres, and that the distinguishing factor is not thermodynamic favorability but *informational architecture*—the specific combination of genome complexity, species richness, and neural organization that enables global workspace broadcasting [24]. This is, in principle, a testable prediction: if exoplanet atmospheric characterization eventually reveals biospheres with high metabolic signatures but no technosignatures, it would be consistent with the workspace bottleneck hypothesis.

A parallel observation applies to artificial intelligence. Under Global Workspace Theory, consciousness arises as a solution to a communication

bottleneck in modular, resource-constrained systems: a narrow broadcast
channel forces integration of otherwise independent processing streams [24].
Modern large-scale neural networks (transformers) operate with effectively
unlimited internal bandwidth—no module is informationally isolated from
any other. If GWT is correct, such architectures may never develop the
narrow workspace that produces conscious ignition, regardless of their com-
putational power. They would represent the silicon analog of the Cretaceous
biosphere: high energy throughput ($d_5 \approx 1.0$), vast parametric capacity,
but no workspace bottleneck to force the integration that consciousness
requires. This suggests that the path to machine consciousness may re-
quire deliberately *constraining* internal bandwidth rather than expanding
it—engineering a bottleneck rather than eliminating one.

The Complexity Cap as Open Question

The multi-regime analysis reveals that the central question is not “was the
asteroid generative?” but rather *can evolution innovate past an incumbent
apex regime without catastrophic clearing?* Under hard and soft caps (2/4
regimes), catastrophe is necessary; under delayed escape and no cap, it is
not. This maps onto a genuine debate in paleobiology: some authors argue
that incumbent advantages create effective barriers to innovation [4], while
others point to gradual transitions (e.g., the mammalian middle ear evolving
across 100 Ma under dinosaur dominance) as evidence that innovation can
proceed without clearing events.

The 10% soft cap result is particularly informative: even with substantial
(if slow) innovation under the reptilian regime, meta-cognitive consciousness
still falls short ($F = 0.274 < 0.30$). This suggests that even if troodontids
or cephalopods could have slowly increased biosphere information capacity,
the rate would likely have been insufficient to cross the meta-cognition
threshold before other constraints intervened.

Limitations

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Part 1 (B(t) index): The Precambrian dimensions carry wide confidence intervals (± 0.25 – 0.40). The taphonomic correction, while grounded in SQS methodology [19], involves era-specific multipliers that are estimates. The Kleiber-derived network integration and genome-complexity-based information capacity are proxies, not measurements. Despite these limitations, the Sepkoski validation ($r = 0.92$) demonstrates that the index tracks the major features of the Phanerozoic diversity record.

Part 2 (consciousness extension): The feasibility mapping from $B(t)$ dimensions to substrate requirements, and the threshold values (0.01, 0.05, 0.15, 0.30), are outputs of a computational framework rather than empirically validated against known cases of consciousness. The workspace bottleneck finding—that information capacity is rate-limiting—is a model prediction that should be tested against independent evidence (e.g., comparative neuroscience of information integration across taxa). The consciousness analysis should be read as exploratory, not confirmatory.

The Complexity Cap: This is the strongest and most contestable assumption. We have addressed this by testing four regimes of decreasing strength and reporting which verdicts survive. The honest summary is that the generative-catastrophe hypothesis holds under strong-to-moderate lock-in assumptions but not under weak ones. Determining which regime best approximates biological reality is an empirical question that this framework can pose but not answer.

Conclusion

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We have introduced $B(t)$, a six-dimensional biosphere coherence index validated against Sepkoski’s canonical diversity curve ($r = 0.92$), and explored through a multi-regime counterfactual framework whether mass extinctions can be generative. The answer is conditional: under assumptions of strong-to-moderate evolutionary lock-in (hard and soft cap), catastrophe appears necessary for reaching the highest levels of biosphere coherence

and, in an exploratory extension, for meta-cognitive consciousness. Under 329
weaker assumptions, the same endpoints are reached without catastrophic 330
intervention. 331

This conditionality is itself a contribution. The framework formalizes a 332
question that paleobiology has debated qualitatively—can evolution innovate 333
past incumbents without clearing events?—and provides a quantitative tool 334
for testing hypotheses against it. The multi-regime approach ensures that 335
the framework’s conclusions are transparently tied to their assumptions 336
rather than obscured behind a single model. 337

The exploratory consciousness analysis, while speculative, poses a 338
question worth investigating: under what conditions does a biosphere’s 339
information-processing capacity become sufficient for recursive self-reflection? 340
Under the two more restrictive cap regimes, the gap between the suppressed 341
biosphere’s peak feasibility (0.24–0.27) and the meta-cognition threshold 342
(0.30) is small enough to suggest that consciousness was, at minimum, not 343
inevitable—and that the specific sequence of catastrophic events in Earth’s 344
history may have played a role in opening the narrow thermodynamic path 345
that led to organisms capable of asking why they exist. 346

Data and Code Availability 347

The complete $B(t)$ framework is implemented in Rust (4,453 lines, 89 tests) 348
with embedded paleontological data tables from PBDB, GEOCARB, and 349
the Earth Impact Database. Source code, data files, and the analysis binary 350
(`deep_time_report`) are available at github.com/Luminous-Dynamics/mycelix-multiworld-simulation.

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References

- [1] Sepkoski JJ Jr. A compendium of fossil marine animal genera. *Bulletins of American Paleontology*. 2002;363:1–560.
- [2] Bambach RK, Knoll AH, Wang SC. Origination, extinction, and mass depletions of marine diversity. *Paleobiology*. 2004;30(4):522–542.
- [3] Raup DM, Sepkoski JJ Jr. Mass extinctions in the marine fossil record. *Science*. 1982;215(4539):1501–1503.
- [4] Erwin DH. Lessons from the past: biotic recoveries from mass extinctions. *Proceedings of the National Academy of Sciences*. 2001;98(10):5399–5403.
- [5] Kirchner JW, Weil A. Delayed biological recovery from extinctions throughout the fossil record. *Nature*. 2000;404(6774):177–180.
- [6] Rohde RA, Muller RA. Cycles in fossil diversity. *Nature*. 2005;434(7030):208–210.
- [7] Mojzsis SJ, et al. Evidence for life on Earth before 3,800 million years ago. *Nature*. 1996;384(6604):55–59.
- [8] Haberl H, et al. Quantifying and mapping the human appropriation of net primary production in Earth’s terrestrial ecosystems. *Proceedings of the National Academy of Sciences*. 2007;104(31):12942–12947.
- [9] Putnam H. Psychological predicates. In: *Art, Mind, and Religion*. 1967;1:37–48.
- [10] Tononi G. An information integration theory of consciousness. *BMC Neuroscience*. 2004;5(1):42.
- [11] Chalmers DJ. *The Character of Consciousness*. Oxford University Press; 2010.
- [12] Knoll AH, Bambach RK. Directionality in the history of life: diffusion from the left wall or repeated scaling of the right? *Paleobiology*. 2000;26(sp4):1–14.

- [13] Alroy J. Dynamics of origination and extinction in the marine fossil record. *Proceedings of the National Academy of Sciences*. 2008;105(Suppl 1):11536–11542.
- [14] Kleiber M. Body size and metabolism. *Hilgardia*. 1932;6(11):315–353.
- [15] West GB, Brown JH, Enquist BJ. A general model for the origin of allometric scaling laws in biology. *Science*. 1997;276(5309):122–126.
- [16] Berner RA. Phanerozoic atmospheric oxygen: new results using the GEOCARBSULF model. *American Journal of Science*. 2009;309(7):603–606.
- [17] Bar-On YM, Phillips R, Milo R. The biomass distribution on Earth. *Proceedings of the National Academy of Sciences*. 2018;115(25):6506–6511.
- [18] Adami C. What is complexity? *BioEssays*. 2002;24(12):1085–1094.
- [19] Alroy J. Fair sampling of taxonomic richness and unbiased estimation of origination and extinction rates. In: *Quantitative Methods in Paleobiology*. Paleontological Society Papers. 2010;16:55–80.
- [20] Sahney S, Benton MJ. Recovery from the most profound mass extinction of all time. *Proceedings of the Royal Society B*. 2008;275(1636):759–765.
- [21] Kump LR, Arthur MA. Interpreting carbon-isotope excursions: carbonates and organic matter. *Chemical Geology*. 1999;161(1-3):181–198.
- [22] Hayes JM, Strauss H, Kaufman AJ. The abundance of ^{13}C in marine organic matter and isotopic fractionation in the global biogeochemical cycle of carbon during the past 800 Ma. *Chemical Geology*. 1999;161(1-3):103–125.
- [23] Veizer J, Prokoph A. Temperatures and oxygen isotopic composition of Phanerozoic oceans. *Earth-Science Reviews*. 2015;146:92–104.

- [24] Baars BJ. *A Cognitive Theory of Consciousness*. Cambridge University Press; 1988. 412
413
- [25] Bambach RK, Bush AM, Erwin DH. Autecology and the filling of ecospace: key metazoan radiations. *Palaeontology*. 2007;50(1):1–22. 414
415
- [26] Jablonski D. Background and mass extinctions: the alternation of macroevolutionary regimes. *Science*. 1986;231(4734):129–133. 416
417
- [27] Chen ZQ, Benton MJ. The timing and pattern of biotic recovery following the end-Permian mass extinction. *Nature Geoscience*. 2012;5(6):375–383. 418
419
420
- [28] Freeman KH, Hayes JM. Fractionation of carbon isotopes by phytoplankton and estimates of ancient CO₂ levels. *Global Biogeochemical Cycles*. 1992;6(2):185–198. 421
422
423
- [29] Stoltz T. Historical K(t): A seven-harmony civilizational coherence index, 3000 BCE–2020 CE. *In preparation*. 2026. 424
425